



Sri Chaitanya IIT Academy.,India.

A.P. T.S. KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT

JEE-MAIN

Date: 02-01-2026

Time: 09.00Am to 12.00Pm

GTM-08

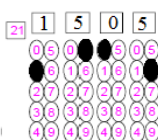
Max. Marks: 300

IMPORTANT INSTRUCTION:

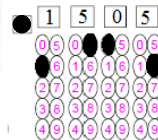
1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
 2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
 3. The test is of **3 hours** duration.
 4. The Test Booklet consists of **75 Questions**. The maximum marks are **300**.
 5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics and Chemistry** having **25 Questions** in each subject and subject having **two sections**.
(I) **Section-I** contains **20 Multiple Choice Questions** with only one correct option.
Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
(II) **Section-II** contains **05 Numerical Value Type Questions**.
- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking



Question Cancelled for Marking

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

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Page 1

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MATHEMATICS : Total Syllabus

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Page 2



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**MATHEMATICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- Let $C_1, C_2, C_3, \dots$ be a sequence of circles with radii $r_1, r_2, r_3, \dots$ such that $r_1 = 1$ (i.e., the equation of C_1 is $x^2 + y^2 = 1$). For every integer $i \geq 1$, the Circle C_i intersects positive X-axis and Y-axis at points A_i and B_i respectively. A point P_i lies on the circle C_i such that if D_i and E_i are the feet of the perpendicular from P_i to the X-axis and Y-axis respectively, then the ratio of their distance from the origin is $OD_i : OE_i = 3 : 4$. Furthermore, there exists a point Q_i on the next circle C_{i+1} such that the triangle $A_i Q_i P_i$ is a right-angled isosceles triangle with the right angle at Q_i (satisfying $A_i Q_i = P_i Q_i$ and $\angle A_i Q_i P_i = \frac{\pi}{2}$). If S_i denotes the area of triangle $A_i Q_i P_i$, then the value of limit $\lim_{n \rightarrow \infty} \left(\sum_{i=1}^n S_i \right)$ is
 1) $\frac{1}{4}$ 2) $\frac{1}{8}$ 3) $\frac{1}{16}$ 4) $\frac{1}{32}$
- Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is
 1) 0 2) 8000 3) 8080 4) 16000
- Let $\int_0^\infty e^{-x^2} dx = a$ and $I_n = \int_0^\infty x^n e^{-x^2} dx, n \in N$ then, the value of I_{98} is equal to
 1) $\frac{196!}{2^{195}(97)!} a$ 2) $\frac{97!}{2^{195}(48)!} a$ 3) $\frac{97!}{2^{97}(48)!} a$ 4) $\frac{195!}{2^{195}(97)!} a$
- Let p, q and r are all different and non-zero real numbers (taken in that order) are in arithmetic progression. If the roots of quadratic equation $px^2 + qx + r = 0$ are a and b such that $a^{-1} + b^{-1}, a + b, a^2 + b^2$ (taken in that order) are in geometric progression, then $\frac{p}{r}$ is equal to
 1) 2 2) 3 3) $\frac{2}{3}$ 4) $\frac{3}{2}$

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Page 3

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5. If $7(A + A^{-1}) = \begin{bmatrix} 12 & 25 & 21 \\ 20 & 23 & 77 \\ -14 & 21 & 21 \end{bmatrix}$ and $\det(A) = -7$, then positive square root of value of

determinant of matrix $(A^{-1} - I_3)^2$ is

- 1) $\frac{19}{5}$ 2) $\frac{18}{7}$ 3) $\frac{21}{8}$ 4) $\frac{23}{7}$

6. Let a_n denote the number of all n -digit numbers formed by the digit 3, 5 or both such that it

- (i) begins with 3
(ii) ends with 3
(iii) contains no two consecutive 3
(iv) contains no three consecutive 5

Then the value of $\frac{a_{15}}{a_{10}}$ is

- 1) $\frac{21}{5}$ 2) $\frac{23}{7}$ 3) $\frac{27}{8}$ 4) $\frac{31}{10}$

7. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as:

$$g(x) = \max(x^2, 4|x| - x^2)$$

The number of points where $g(x)$ is non-differentiable is a and the number of points where $g(x)$ is discontinuous is equal to b . The value of $a+b$ is equal to:

- 1) 0 2) 1 3) 2 4) 3

8. The set of all real values of ' a ' for which both the roots of the equation $x^2 - 1 = 0$ lie between the roots of the equation $x^2 + (3a - a^2)x - 3a^3 = 0$ is equal to

- 1) $(-\infty, -1)$ 2) $\left(-1, -\frac{1}{3}\right) \cup (1, \infty)$ 3) $(1, \infty)$ 4) $\left(-\frac{1}{3}, 1\right)$



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341

RANK

1

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2023

VAVILALA

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RANK 1

341

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1

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722

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1



9. If the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ intersects the line

$$3\beta^2 x + 3(1 - 2\alpha)y + z = 3 = -\frac{1}{2} \left\{ 6\alpha^2 x + 3(1 - 2\beta)y + 2z \right\},$$

then point $(\alpha, \beta, 1)$ lie on the plane L . Which of the following equation CANNOT be the equation of plane L ?

- 1) $2x - y + z = 4$ 2) $x + y - z = 2$ 3) $x - 2y = 0$ 4) $2x - y = 0$

10. In a $\triangle ABC$, r_1, r_2 and r_3 denote the exradii opposite to vertices A, B and C respectively.

If $r_1 = 2r_2 = 3r_3$ then, the value of $25(\sin B + \sin C)(\cos B + \cos C)$ is equal to

- 1) 31 2) 47 3) 49 4) 53

11. The value of $\int_{-20\pi}^{20\pi} [\sin x + \cos x] dx$ is k . Then, the value of $||[k]||$ is equal to _____. (where

$[.]$ represents greatest integer function)

- 1) 71 2) 89 3) 61 4) 63

12. The area enclosed between the curves $y^2 = 4x$ and $x^2 = 4y$ inside the square formed by the lines $x = 1, y = 1, x = 4, y = 4$ is equal to A units square, then the value of $3A$ is equal to _____.

- 1) 21 2) 11 3) 35 4) 48

13. Let $\alpha = e^{i\frac{2\pi}{11}}, \lambda = \alpha^6, \mu = \alpha^7$, and $\beta = \alpha^2$.

Let the complex number Z be defined as: $Z = \operatorname{Re} \left(\sum_{k=1}^5 \lambda^k \right) + \prod_{k=1}^{10} (\mu - \beta^k) + \prod_{k=1}^{10} (i - \beta^k)$

If $Z = n + mi$ (where $n, m \in \mathbb{R}$), match the entries in List-I with their correct values in List-II





	List-I		List-II
(P)	The value of $\operatorname{Re}\left(\sum_{k=1}^5 \lambda^k\right)$	(1)	25
(Q)	The value of $\prod_{k=1}^{10} (\mu - \beta^k)$	(2)	0
(R)	The value of $\prod_{k=1}^{10} (i - \beta^k)$	(3)	$-\frac{1}{2}$
(S)	The value of $50(m+n)$	(4)	i

1) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$ 2) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

3) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$ 4) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

14. Number of solutions of the equation $\sin^{-1}(x-1)^2 + \cos^{-1}\left(\frac{1}{x^2-1}\right) = \tan^{-1}x$ is

(considering only the principal values of inverse trigonometric function)

1) 1 2) 2 3) 3 4) 4

15. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} =$

1) $\frac{2}{3}$ 2) 1 3) $\frac{5}{3}$ 4) $\frac{7}{3}$

16. From a point $P(1,2)$ pair of tangents are drawn to a hyperbola ' H ' in which one tangent to each arm of hyperbola. Equation of asymptotes of hyperbola H are $\sqrt{3}x - y + 5 = 0$ & $\sqrt{3}x + y - 1 = 0$, then eccentricity of ' H ' is

1) $\sqrt{7}$ 2) $\frac{2}{\sqrt{3}}$ 3) $\sqrt{2}$ 4) $\sqrt{3}$

17. Let ' k ' be sum of all ' x ' in the interval $[0, 2\pi]$ such that $3\cot^2 x + 8\cot x + 3 = 0$ Then k is equal to

1) 5π 2) 4π 3) 3π 4) 2π

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Page 6



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341

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1

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VAVILALA
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18. Let $\vec{a} = 2\hat{i} + 3\hat{j}$ and $\vec{b}$ be any vector in xy -plane perpendicular to $\vec{a}$. If a vector $\vec{c}$ in the same plane have lengths of projections along $\vec{a}$ and $\vec{b}$ equal to $\frac{5}{\sqrt{13}}$ and $\frac{3}{\sqrt{13}}$, then $\vec{c} =$

- 1) $\frac{2}{13}\hat{i} + \frac{3}{13}\hat{j}$ 2) $\frac{9}{13}\hat{i} + \frac{19}{13}\hat{j}$ 3) $\frac{19}{13}\hat{i} + \frac{9}{13}\hat{j}$ 4) $\frac{19}{13}\hat{i} + \frac{8}{13}\hat{j}$

19. $f(x)$ is a polynomial of 6th degree and $f(x) = f(2-x) \forall x \in R$. If $f(x) = 0$ has 4 distinct real roots and two real and equal roots, then sum of roots of $f(x) = 0$ is

- 1) 8 2) 7 3) 6 4) 5

20. Let $f(x) = \int_0^x \frac{t}{1+t^4} dt$.

Statement-I: The equation $f(x) = \frac{\pi}{8\sqrt{2}}$ has exactly one real solution.

Statement-II: $\int_0^{\infty} \frac{t}{1+t^4} dt = \frac{\pi}{4\sqrt{2}}$

Options

- 1) Both Statements I and Statement II are correct.
2) Statement I is correct but Statements II is incorrect.
3) Both Statement I and Statements II are incorrect.
4) Statement I is incorrect but Statement II is correct.

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Through the vertex A of the parabola $\left(\frac{z-\bar{z}}{2i}\right)^2 = 2(z+\bar{z})$ where $z = x + iy$ ($i = \sqrt{-1}$) two chords AP and AQ are drawn and the circles on AP and AQ as diameters intersect in R. If α, β be the angles made with the x -axis by the tangents at P & Q on the parabola and γ be the angle between AR and x -axis then find the absolute value of $(\cot \alpha + \cot \beta) \cot \gamma$

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Page 7



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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

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341

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22

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22. A bag contains 3 red, 2 white and 2 black balls. Two balls are drawn at random and none of them is found to be a white ball. The probability that both balls are red is $\frac{a}{b}$ (where a, b are coprime) then $b - a$ is equal to
23. If $16 + \sin^{12} x + \cos^{11} x - \sec^2 y - 9 \operatorname{cosec}^2 y \geq \sec^8 z - 2 \sec z + \log_2(5 - \cos^2 x)$ and $x, y, z \in [0, 2\pi]$ then, number of possible ordered triplets (x, y, z) are
24. If $f(x) = [x]\{x\}$ and $g(x) = |f(x)|$, then the sum of the minimum values of the magnitudes of real number α, β such that $f(\alpha) = 1$ and $g(\beta) = 1$ is equal to ($[.]$ is G.I.F $\{\}$ is F.P.F)
25. The value of limit $\lim_{x \rightarrow 0} \frac{\sin(3x+a) - 3\sin(2x+a) + 3\sin(x+a) - \sin a}{x^3}$ is $-\cos\left(\frac{3a}{\lambda}\right)$.
Then value of λ is (where $\lambda \in \mathbb{N}$)

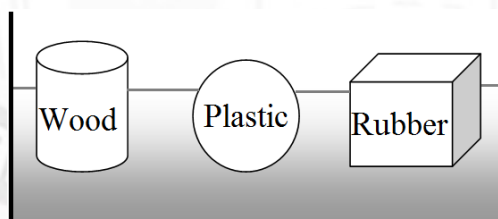


**PHYSICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. A student, intending to catch a bus on the bus stand, finds that the bus starts and accelerates away from him/her with uniform acceleration $a = 2\text{ms}^{-2}$ when he is 16 m far from the bus. He runs fast enough with a uniform speed so as just to catch the bus. The minimum speed with which the student must run, is
 1) 2.0ms^{-1} 2) 4.0ms^{-1} 3) 8.0ms^{-1} 4) 16.0ms^{-1}
27. An electric kettle consists of two filaments F_1 and F_2 of different resistances R_1 and R_2 respectively connected in parallel with separate switches. With a certain fixed amount of water in the kettle pot to prepare the tea, when only F_1 is switched on, it takes 4 minute to boil the water. When F_1 and F_2 both are switched on simultaneously, it takes 3 minute to boil the same amount of water up to the same temperature. The time taken to boil the same amount of water when only F_2 is switched on (assume that entire heat radiated by filaments is used in boiling the water starting from the same room temperature in each case) is
 1) 7.0 min 2) 8.4 min 3) 9.6 min 4) 12.0 min
28. A wooden solid cylinder (density 0.80 g/cm^3), a plastic ball (density 0.90 g/cm^3) and a rubber cube (density 0.95 g/cm^3) all of equal mass M float partly immersed in a trough containing water (density 1.00 g/cm^3) as shown below (ignore any tilt during floatation)



You may conclude that

- 1) The buoyant force is largest on the wooden cylinder
- 2) The buoyant force is largest on the Plastic ball
- 3) The buoyant force is largest on the rubber cube
- 4) The buoyant force is equal on all the three

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Page 9



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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

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341

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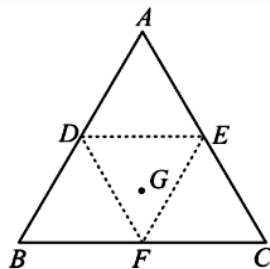


29. Two vertical object pins are located on the principal axis of a thin lens at distances of $x_1 = 9.0\text{cm}$ and $x_2 = 18.0\text{cm}$ from the lens on the two sides of the lens. The images of the two pins formed by the lens are at the same position (overlap). The focal length and the nature of the lens are
- 1) 15 cm, convex
 - 2) 12 cm, convex
 - 3) 15 cm, concave
 - 4) 12 cm, concave
30. Given below are two statements:
- Statement I: Electromagnetic waves carry energy as they travel through space and this energy is equally shared by the electric and magnetic fields.
- Statement II: When electromagnetic waves strike a surface, a pressure is exerted on the surface. In the light of the above statements, choose the most appropriate answer from the options given below:
- 1) Statement I is incorrect but Statement II is correct
 - 2) Both Statement I and Statement II are correct.
 - 3) Both Statement I and Statement II are incorrect.
 - 4) Statement I is correct but Statement II is incorrect.
31. Knowing that the atomic masses of Al and Mg are respectively $^{25}_{13}\text{Al} = 24.990432\text{ u}$ and $^{25}_{12}\text{Mg} = 24.985839\text{ u}$ while electron mass is often expressed as $m_e = 0.511\text{ MeV}$, the Q value (energy liberated) of the β decay nuclear reaction $^{25}_{13}\text{Al} \rightarrow ^{25}_{12}\text{Mg} + e^+ + \nu$ in MeV is
- 1) 4.278
 - 2) 3.767
 - 3) 3.256
 - 4) 931.478
32. Liquid droplets store a given amount of potential energy per unit surface area, due to their surface tension. When two identical, nearly spherical liquid droplets coalesce on a certain type of surface, part of this energy can be converted into upward kinetic energy, causing the coalesced droplet to jump. Assuming the conversion is 100% efficient, how does the maximum height h depend on the radius r of the initial droplets?
- 1) $h \propto r$
 - 2) $h \propto r^{1/2}$
 - 3) $h \propto r^{-1/2}$
 - 4) $h \propto r^{-1}$





33. An equilateral triangle ABC is cut from a thin solid sheet of wood. (See figure) D, E and F are the mid-points of its sides as shown and G is the centre of the triangle. The moment of inertia of the triangle about an axis passing through G and perpendicular to the plane of the triangle is I_0 . If the smaller triangle DEF is removed from ABC , the moment of inertia of the remaining figure about the same axis is I . Then:



consider following two statements in the light of above problem

Statement-1 Moment of inertia $I = (15I_0) / 16$

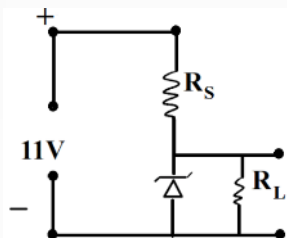
Statement-2 Moment of inertia of the triangle DEF about centre G will be 16 times less as compared to that of ABC about G and symmetry considerations suggest that moment of inertia of remaining three triangular shapes ADE , DBF and EFC will be same. Finally we can super impose moment of inertia of above four triangular section to get the result.

- 1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - 2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 - 3) Statement-1 is True, Statement-2 is False.
 - 4) Statement-1 is False, Statement-2 is True.
34. Rank the vernier callipers/screw gauge in the order of their increasing precision :
- (1) Screw gauge having main scale division 1mm and no. of circular division = 200.
 - (2) Screw gauge having main scale division 0.5 mm and no. of circular division = 400.
 - (3) Vernier callipers having main scale division = 1mm such that 9 MSD = 10 VSD.
 - (4) Vernier callipers having main scale division = 1mm such that 4 MSD = 5 VSD.
- 1) $4 < 2 < 3 < 1$ 2) $2 < 4 < 1 < 3$ 3) $4 < 3 < 2 < 1$ 4) $4 < 3 < 1 < 2$

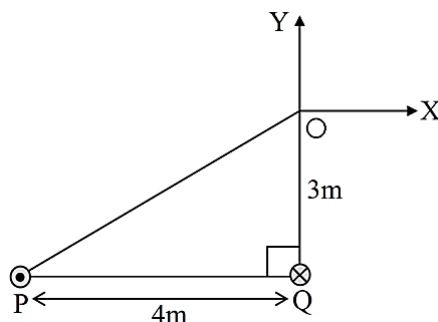




35. In the circuit shown, the Zener diode is an ideal one with breakdown voltage of 5.0 volt. The values of the resistances are $R_s = 10 \text{ k}\Omega$ and $R_L = 1 \text{ k}\Omega$. The current through the resistances, when the supply voltage is 11.0 V, is



- 1) 0.6 mA through R_s and 5.0 mA through R_L
 - 2) 1.0 mA through R_s and 1.0 mA through R_L
 - 3) 1.1 mA through R_s and no current through R_L
 - 4) no current through R_s and 11 mA through R_L
36. Two infinitely long straight parallel wires perpendicular to the plane of the paper are 4 m apart. One of the wires, P carries current I out of the plane of the paper and the other, Q carries the current I into the plane of paper. The magnetic field B at the origin O of the coordinate system with x and y axes as perpendicular and parallel to PQ, respectively, is [Given OP = 5 m and OQ = 3 m]

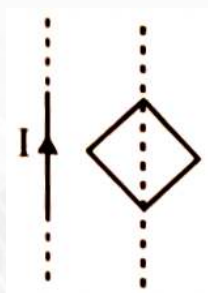


- 1) $\left(\frac{\mu_0}{4\pi}\right) 2I \left\{ \frac{4}{25} j + \frac{16}{75} i \right\}$
- 2) $\left(\frac{\mu_0}{4\pi}\right) 2I \left\{ \frac{4}{25} j - \frac{16}{75} i \right\}$
- 3) $\frac{\mu_0}{4\pi} I \left\{ \frac{4}{25} j - \frac{16}{25} i \right\}$
- 4) Zero





37. A certain substance, with a dielectric constant $k = 2.5$ and the dielectric strength $E = 1.8 \times 10^7 \text{ N/C}$, completely fills the space between the plates of a parallel plate capacitor (with circular plates) of capacitance $C = 72.0 \text{ nF}$. The minimum diameter of the circular plates, to ensure that the capacitor can withstand a potential difference of $V = 4.0 \text{ kV}$, is
- 1) 12 cm 2) 24 cm 3) 48 cm 4) 96 cm
38. A square loop of conducting wire is placed symmetrically near a long straight current carrying wire as shown. Match the statements in Column-I with the corresponding results in Column-II.



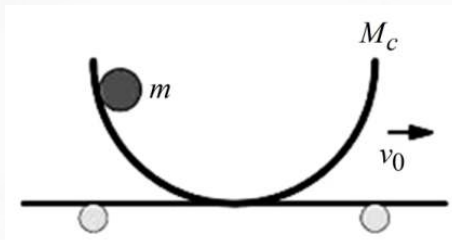
	Column-I		Column-II
(A)	If the magnitude of current I is increased	(P)	Current will induce in clockwise direction in the loop
(B)	If the magnitude of current I is decreased	(Q)	Current will induce in anticlockwise direction in the loop
(C)	If the loop is moved away from the wire	(R)	Wire will attract the loop
(D)	If the loop is moved towards the wire	(S)	Wire will repel the loop
		(T)	Loop will rotate when current changes.

- 1) A-Q, S; B-P,R; C-P; D-Q 2) A-Q, R; B-P,R; C-P; D-Q
- 3) A-Q; B-P; C-P; D-Q 4) A-Q, S; B-P,R; C-P,R; D-Q, S





39. A ball of mass m rolls in a bowl bolted on a cart. The bowl and cart move together and have a combined mass M_c , as shown in the figure. Initially, the cart moves to the right with speed v_0 with respect to the ground, while the ball is at rest with respect to the cart. The ball then slides down to the bottom of the bowl. When it reaches the bottom of the bowl, it moves with velocity v_b with respect to the cart. At this instant, how fast is the cart moving with respect to the ground?



- 1) $v_0 - v_b$ 2) $v_0 - \frac{m}{M_c} v_b$ 3) $\frac{m_c v_0 - m v_b}{M_c + m}$ 4) $\frac{(m + M_c) v_0 - m v_b}{m + M_c}$
40. Assertion:- Two particles of mass 1 kg and 3 kg move towards each other under their mutual force of attraction. No other force acts on them. When the relative velocity of approach of two particle is 2 m/sec, their centre of mass has velocity of 0.5 m/sec. When the relative velocity of approach increases to 3 m/sec, the velocity of the centre of mass decreases to 0.25 m/sec

Reason:- The total kinetic energy as seen from ground is $\frac{1}{2} \mu V_{rel}^2 + \frac{1}{2} M V_{c.m}^2$ and here μ is

reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$ and M is total mass $M = (m_1 + m_2)$ and in absence of

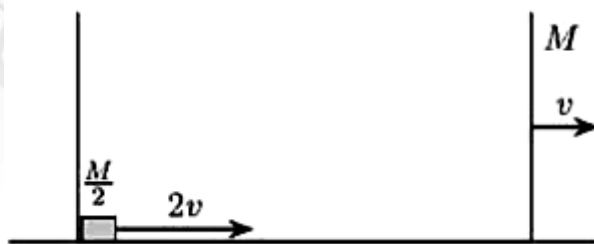
external force, total mechanical energy remains conserved

- 1) Assertion is true, Reason is true 2) Assertion is false, Reason is true
3) Assertion is false, Reason is false 4) Assertion is true, Reason is false
41. Let us consider two statements and based upon those, choose the most suitable option
- Statement-1 A conducting body can't be charged beyond a particular limiting value of charge
- Statement-2 Medium surrounding the conducting body has a finite value of dielectric strength, therefore charging beyond the particular value results in breakdown of the medium.





- 1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- 2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
- 3) Statement-1 is True, Statement-2 is False.
- 4) Statement-1 is False, Statement-2 is True.
42. One mole of an ideal monoatomic gas, contained in a cylinder fitted with movable piston, is originally at P_1 , V_1 , and $T_1 = 27^\circ\text{C}$. The gas is slowly heated. Initially 8.31 watt-hour of energy is added to it; at the same time it is allowed to expand at constant pressure to a new state P_1 , V_2 and T_2 . Choose incorrect option
- 1) Value of T_2 is 1740°C
- 2) Work done by the gas is 1440 R joule
- 3) Internal energy of the gas increases by 2160 R joule
- 4) $\frac{V_2}{V_1} = 5.8$
43. A box of mass M is sliding to the right with velocity v on a frictionless table. A small puck of mass $M/2$ slides frictionlessly inside the box. Initially, the puck is at the left wall of the box, with a rightward velocity of $2v$ with respect to the table. After a time T , the puck collides elastically with the right wall of the box. How much longer will it take until the puck hits the left wall of the box again?

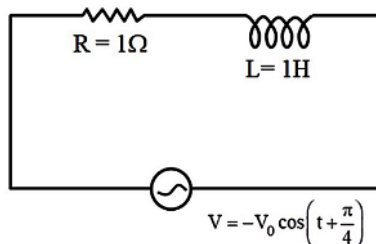


- 1) $T/2$ 2) T 3) $2T$ 4) $3T$





44. The instantaneous voltage drop V_R across resistor in circuit is



1) $V_R = \frac{-V_0}{\sqrt{2}} \cos t$

2) $V_R = V_0 \cos t$

3) $V_R = \frac{-V_0}{\sqrt{2}} \cos\left(t + \frac{\pi}{4}\right)$

4) $V_R = \frac{-V_0}{\sqrt{2}} \cos\left(t - \frac{\pi}{4}\right)$

45. Following is a problem based upon YDSE / modified YDSE set up and we perform few activities (independently of one another) as described.

1. The screen is moved towards the plane of slits
 2. Separation between two slits is increased
 3. White light is used instead of monochromatic
 4. Set up is immersed in some liquid
- 1) In (1) fringe width increases and angular width remains same
 - 2) In (3) central fringe is white surrounded by few coloured fringes
 - 3) In (4) fringe width remains unaffected
 - 4) No effect whatsoever is seen in (2)

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. If P represents radiation pressure, E represents radiation energy striking per unit area per unit time and c represents speed of light then find the value of $x + y + z$ such that

$P^x E^y c^z$ is dimensionless. (x, y, z are integers and we have to find smallest possible positive solutions for $x+y+z$)

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Page 16



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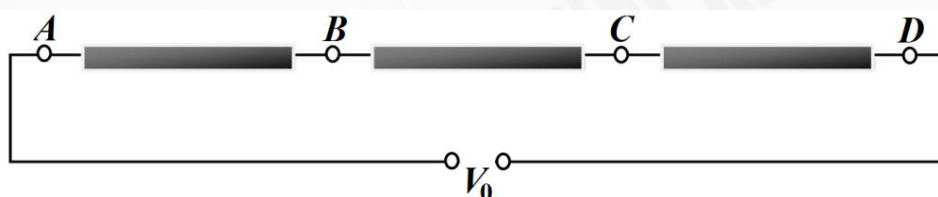
72

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47. A thermometer graduated according to a linear scale reads a value x_0 when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in $^{\circ}\text{C}$, if this thermometer in the contact with the object reads $x_0/2$?
48. Three resistors are connected in series with a battery of internal voltage $V_0 = 62V$ and internal resistance $R_b \approx 0$ as shown in the figure, then measurements are carried out with a single voltmeter. The results of the first three measurements are: $V_{AB} = V_{BC} = V_{CD} = 20V$. Find the absolute value of difference in measured voltages across AC and AD , that is $V_{AD} - V_{AC}$



49. A solid non conducting sphere of radius R has radially symmetrical distribution of charge over its volume. The ratio of Intensity of electrostatic field at a distance $R/2$ and at a point on the surface is 0.125. If it is known that volume charge density is directly proportional to n power of radial distance r then find the value if n
50. A 20 cm long string having a mass of 1.0 g, is fixed at both the ends. The tension in the string is 0.5 N. The string is set into vibration using an external vibrator of frequency 100 Hz. Find the separation (in cm) between the successive nodes on the string.



**CHEMISTRY****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. Which one of the following acids does not exhibit optical isomerism?
- Lactic acid
 - Tartaric acid
 - Maleic acid
 - α -amino acids
52. The reaction product $A \xrightarrow{k} \text{Product}$, is zero order while the reaction $B \xrightarrow{k} \text{Product}$, is first order reaction. For what initial concentration of A are the half lives of the two reaction equal? (Rate constant value are same for both two reaction).
- $(\log_e 4)M$
 - $2M$
 - $\log 2M$
 - $\ln 2M$
53. In which of the following arrangement, the order is according to the property indicated against it:
- $I < Br < F < Cl$: Increasing EA
 - $Li < Na < K < Rb$: Increasing metallic radius
 - $B < C < N < O$: Increasing first ionisation energy
 - $Al^{+3} < Mg^{+2} < Na^{+} < F^{-}$: Increasing ionic size
- I, II and III
 - I & III Only
 - I, II and IV
 - I & II Only
54. **Statement-1:** Glucose gives a reddish-brown precipitate with Fehling's solution. because
- Statement-2:** Reaction of glucose with Fehling's solution gives CuO and gluconic acid.
- Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is False.
 - Statement-1 is False, Statement-2 is True.

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Page 18

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55. A weak acid HA after treatment with 12 ml of 0.1M strong base BOH has pH = 5. At the end point, the volume of same base required is 26.6 ml. K_a of acid is :

[Given: $\log\left(\frac{1.2}{1.46}\right) = -0.085$]

- 1) 8.2×10^{-6} 2) 8.2×10^{-5} 3) 8.2×10^{-4} 4) 8.2×10^{-7}
56. The photon radiated from hydrogen corresponding to 2^{nd} line of Lyman series is absorbed by a hydrogen like atom x in 2^{nd} excited state. As a result the hydrogen like atom 'x' makes a transition to n^{th} orbit. Then

1) $x = He^+, n = 4$ 2) $x = Li^{++}, n = 6$ 3) $x = He^+, n = 6$ 4) $x = Li^{++}, n = 9$

57. Which of the following property is not related to solution of rosy red complex of Ni^{2+} and dimethyl glyoximate ligand?

- 1) Geometry around Ni is square planar and diamagnetic in nature
2) Complex is stabilized by internal hydrogen bonding
3) it is an organometallic complex
4) Five member chelate ring is formed when each molecule of dimethyl glyoximate is bonded with metal ion.

58. An organic compound (X) is subjected to reductive ozonolysis to give a product (A) which on heating with dil. $NaOH$ gives another product (B). Compound (B) on heating with sodium hypoiodite followed by acidification produces (C). Compound (C) on treatment with $SOCl_2$ and then with diazomethane followed by hydrolysis in acidic medium gives a product which on heating gives 1, 2-dimethyl cyclopentene.

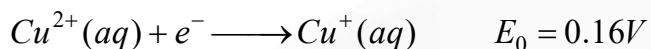
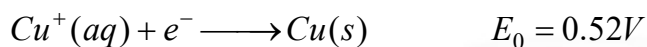
Organic compound (X) is

- 1) 1,2-dimethyl cyclopentene 2) 1,2-dimethyl hexene
3) 1,2-dimethyl cyclohexene 4) 2,5-dimethyl-3-hexene

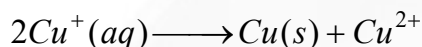




59. Given the standard potential for the following half-cell reaction at 298K,



Calculate the $\Delta G^0(kJ)$ for the reaction,



- 1) -34.740 2) -65.720 3) -69.480 4) -131.440

60. Match List – I with List- II

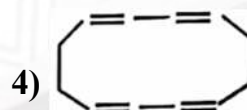
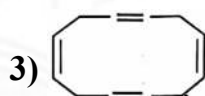
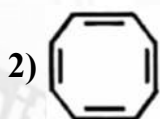
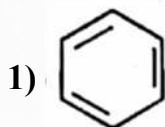
List – I		List – II	
Electronic configuration of elements		$\Delta_i H$ in kJ mol^{-1}	
(a)	$1s^2, 2s^2$	(i)	801
(b)	$1s^2, 2s^2, 2p^4$	(ii)	899
(c)	$1s^2, 2s^2, 2p^3$	(iii)	1314
(d)	$1s^2, 2s^2, 2p^1$	(iv)	1402

Choose the most appropriate answer from the options given below:

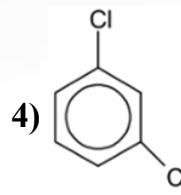
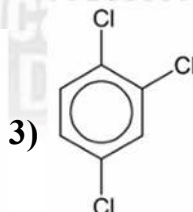
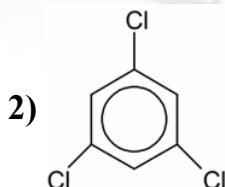
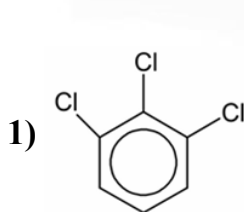
1) (a)-(ii), (b) - (iii), (c) - (iv), (d) - (i) 2) (a)-(i), (b) - (iii), (c) - (iv), (d) - (ii)

3) (a)-(i), (b) - (iv), (c) - (iii), (d) - (ii) 4) (a) - (i), (b) - (iv), (c) - (ii), (d) - (iii)

61. An organic hydrocarbon on oxidative ozonolysis produces oxalic acid and butanedioic acid. Its structure is



62. Which of the following has maximum dipole moment ?

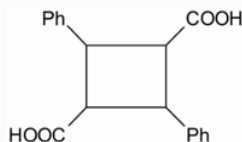




63. At 37°C , the osmotic pressure of blood is 7.65 atm. How much glucose should be used per litre for an intravenous injection that is to have the same osmotic pressure as blood.

- 1) 0.3 g 2) 3×10^{-3} g 3) 0.54 g 4) 54 g

64. How many geometrical isomers are possible for the following compound ?



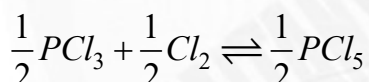
- 1) 5 2) 6 3) 7 4) 8

65. When N_2^+ is formed from N_2 , bond order

and when O_2^+ is formed from O_2 bond order

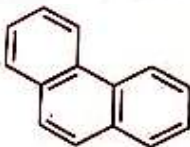
- 1) increases, increases 2) decreases, decreases
3) increases, decreases 4) decreases, increases

66. A reaction at equilibrium involving 2 moles each PCl_5 , PCl_3 and Cl_2 is maintained at 250°C and a total pressure of 12 atm. The value of K_p for the reaction



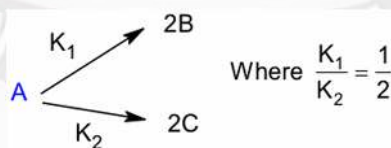
- 1) 2 atm 2) 4 atm 3) 0.25 atm 4) $2\text{ atm}^{-1/2}$

67. Number of resonating structures of phenanthrene are



- 1) 4 2) 6 3) 5 4) 3

68. For a hypothetical elementary reaction:



Initially only 2 moles of A are present. The total no. of moles A, B and C at the end of 75% reaction are

- 1) 2 2) 3 3) 4 4) 3.5



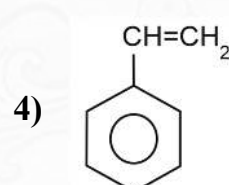
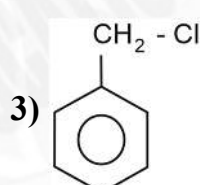
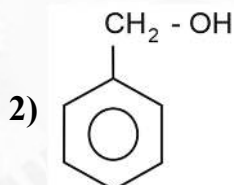
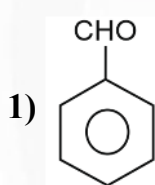
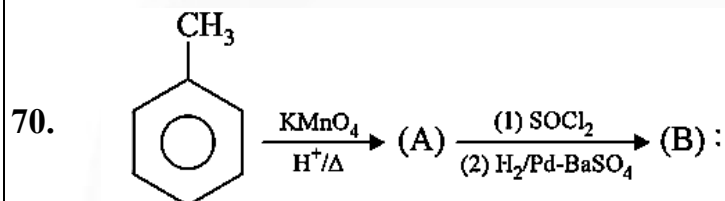


69. Given below are two statements:

Statement 1 : Aqueous solution of YbSO_4 is a strong reductant.

Statement 2 : Tb in TbO_2 has half-filled f -subshell and acts as an oxidant.

- 1) Statement 1 is correct but statement 2 is incorrect
- 2) Statement 1 is incorrect but statement 2 is correct
- 3) Both statement 1 and statement 2 are correct
- 4) Both statement 1 and statement 2 are incorrect



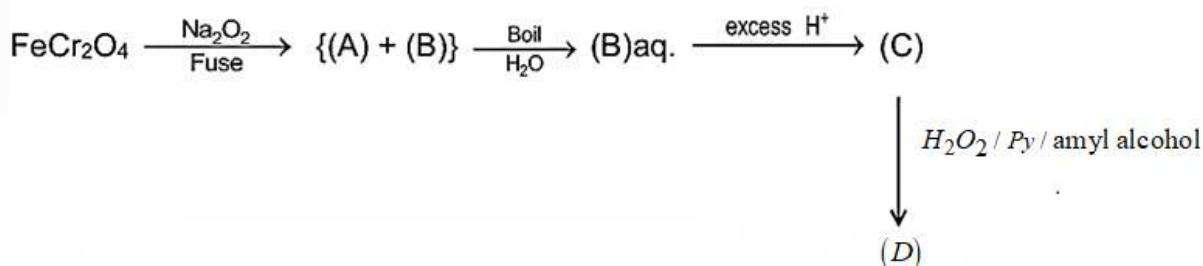
SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. Chromite ore is processed throughout the following sequence :

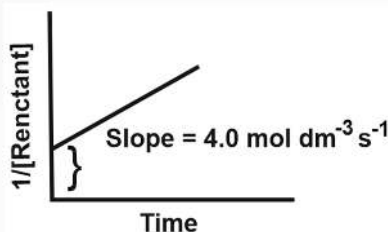


In this sequence, the oxidation state of transition metal in product (D) is





72. The data of a chemical reaction is plotted as $\frac{1}{[\text{Reactant}]}$ vs time and the plot is a straight line with intercept $4.0 \times 10^2 \text{ mol}^{-1} \text{ dm}^3$ and slope $4.0 \text{ mol dm}^3 \text{ s}^{-1}$ as shown in figure. Calculate the half-life of the reaction (in Seconds)



73. Find the enthalpy change ΔH for heating 1 mole of an ideal gas from 300 K to 400 K. The molar heat capacity at constant pressure is given by:
 $C_p = 20.0 + 0.01T$ (in $\text{J K}^{-1} \text{ mol}^{-1}$)
 (Assume ideal behaviour.) (Round off your result to nearest integer)
74. Solubility of $\text{Pb}(\text{OH})_2(s)$ in pure water is $10^{-4} M$. How many times the solubility of $\text{Pb}(\text{OH})_2$ will increase in buffer of $\text{pH} = 10$? (Report nearest integer value).
75. A three-part problem is given below. In each case, consider only major product(s).
 (A) Methylcyclohexene undergoes reductive ozonolysis. The resulting compound is then treated with dilute aqueous NaOH followed by heating. The number of major product(s) formed is X .
 (B) A mixture of formaldehyde and benzaldehyde is treated with concentrated NaOH . The number of major product(s) formed is Y .
 (C) 1,2-Dimethylcyclohexene reacts with meta-chloroperoxybenzoic acid to give product(s). Count only structurally distinct major product(s) (do not count enantiomers separately). The number of major product(s) formed is Z .
 The numerical answer should be entered as the three-digit number XYZ , obtained by concatenating the values of X , Y , and Z in that order.

If a problem of this type yields:

$$X = a, Y = b, Z = c$$

then the numerical answer to be entered is: abc

(This is an illustration of format only and does not correspond to the values in this question)





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ALL INDIA OPEN
CATEGORY RANKS

31

BELOW
500
ALL INDIA OPEN
CATEGORY RANKS

95

BELOW
10
ALL INDIA CATEGORY
RANKS COUNT

10

BELOW
100
ALL INDIA CATEGORY
RANKS COUNT

98

BELOW
1000
ALL INDIA CATEGORY
RANKS COUNT

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CATEGORY RANKS

205

BELOW
1000
ALL INDIA CATEGORY
RANKS COUNT

745

**NUMBER OF
QUALIFIED RANKS**
4,212

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